Ayush Gupta

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Summary

Software Engineer focused on **backend** architecture, **Machine Learning** infrastructure, and distributed systems. Active **open-source contributor** to Kubeflow. Experienced in optimizing system latency, building full-stack applications, and writing production-ready, heavily tested code.

Experience

Mar 2026 - PRESENT

Open Source Contributor

Kubeflow

- Katib (AutoML): Fixed HuggingFace trainer startup crashes by implementing a robust JSON parser for training_parameters. Ensured backward compatibility for empty payloads and added comprehensive pytest coverage for edge cases (PR #2656).
- Docs-Agent: Contributed to the documentation agent's architecture, optimizing backend retrieval and data handling to improve response reliability (PR #186).

Jan 2026 - PRESENT

Contributor

DSC Winter of Code

- Contributed to open-source repositories, focusing on bug fixes, code maintainability, and passing automated CI/CD checks during technical reviews

Education

2024 - 2028

VIT Bhopal University, Bhopal

B.Tech In Computer Science & Engineering

CGPA - 8.45

2022 - 2023

Brij Kunwar Devi Aldrich Public School (CBSE)

Class XII

Percentage - 74%

Skills

- Data Structures & Algorithms
- Object-Oriented Programming (OOP)
- System Architecture
- Python
- C++
- SQL
- JavaScript
- Java
- Relational Databases
- GitHub Actions
- CI/CD Pipeline
- Pytest
- Next.js
- React
- Node.js
- Kubeflow
- TensorFlow
- PyTorch
- OpenCV
- Pandas
- HuggingFace

Projects

AI Surveillance System

Mar 2026 – Apr 2026

- Engineered a deep learning pipeline combining spatial (**CNN**) and temporal (**RNN**) video analysis to achieve real-time anomaly detection in **CCTV streams**.
- Designed a parallel streaming and inference pipeline using **Python multiprocessing**, ensuring low-latency video processing for continuous surveillance.
- Implemented computer vision models for automated activity recognition, significantly reducing the manual effort required for threat identification.

HerWay – Predictive Safe Route Navigation System

Jan 2026 – Apr 2026

Project Link

- Built an AI-driven safety navigation system that identifies safer travel routes by analyzing real-time factors like location data, crime rates, and proximity to emergency services using machine learning models.
- Developed real-time tracking and emergency response features, enabling SOS alerts and live location sharing to improve user safety during travel scenarios.
- Designed and integrated a scalable mobile architecture (Android + backend services) with location-based APIs and data pipelines to support continuous route evaluation and user interaction.

Sunrise Hotel Website

Aug 2025 – Nov 2025

Project Link

- Built a production-ready, full-stack hotel booking platform using **Next.js (TypeScript) and Tailwind CSS**, optimizing for fast load times and responsive UX.
- Architected a secure backend using **Supabase**, implementing **SMTP**-based email authentication to improve user management and reliability.
- Deployed the application via Vercel with modern **CI/CD** workflows, ensuring high availability, consistent uptime, and seamless updates.

Yoga Pose Detection Engine

Jul 2025 – Nov 2025

Project Link

- Implemented **computer vision-based keypoint detection** models to classify human poses in real-time, greatly improving posture accuracy detection.
- Mapped dynamic body **joint angles** to predefined pose templates to deliver automated, instant feedback for user posture correction.
- Optimized the video frame processing loop and integrated lightweight pose models to guarantee efficient inference and a smooth user experience.

Achievements

Open Source Architect - Kubeflow

Successfully merged core architecture and bug-fix pull requests into Kubeflow (**Katib and Docs-Agent**), directly impacting a globally utilized machine learning infrastructure platform

DSC Winter of Code 2026

Selected from a highly competitive applicant pool as an open-source contributor to maintain and scale community-driven repositories.

#T20DSAChallenge Competitor

Completed an intensive data structures and algorithms challenge, demonstrating advanced proficiency in Stacks, Queues, and complex algorithmic problem-solving.

Technical Certifications

Applied Machine Learning in Python (University of Michigan / Coursera) & *Introduction to Generative AI* (Google Cloud Skills Boost).

Responsibility

MATRIX Club, VIT Bhopal University

Core Member - Technical Team

Collaborate with team members to organize coding competitions and technical workshops. Mentor junior students in Python programming and Data Structures.

TechnoMech Club, VIT Bhopal University

Core Member - Technical Team

Architected and deployed internal campus tools for 5,000+ users as a Core Technical Member, improving system load times by 82% and establishing strict version control practices.